Xu Zhang

Email: zx0417@connect.hku.hk
Homepage: https://deadworm.github.io/

Education

The University of Hong Kong, Hong Kong, China *Sep. 2020 - Aug. 2024*
Doctor of Philosophy, Physics
PhD physics course names and grades:
PHYS8552 - 2021 -1 Physics of Quantum Liquids (A-)
PHYS8750 - 2021 -1 Nanophysics (A-)
PHYS8550 - 2021 -2 Graduate Statistical Mechanics (A-)
PHYS8351 - 2021 -2 Graduate Quantum Mechanics (A+)
PhD thesis: Electromagnetic response in two dimensional flat band systems
Thesis Advisor: Zi Yang Meng

Nankai University, Tianjin, China *Sep. 2015 - Jun. 2019*
Bachelor of Science, Physics
GPA: 90.06/100 (Ranking: 5/75)
Thesis Advisors: Jiangping Hu, Liang Jin

Employment & Exchange Experience

Ghent University, Gent, Belgium *Sep. 2024 - Present*
PostDoc, Advisors: Nick Bultinck, Jutho Haegeman and Frank Verstraete
Theory and simulation on strongly correlated systems

University of Michigan, MI, USA *Feb. 2023 - Mar. 2023*
Junior Short Term Visitor, Host: Kai Sun
Nonlinear Hall effect and fractional Chern insulator simulation by Quantum Monte Carlo

Institution of Physics, Beijing, China *Jun. 2019 - Sep. 2020*
Research Assistant, Advisors: Jiangping Hu, Zi Yang Meng
Density Matrix Renormalization Group computation for 1D copper-oxygen chain and Determinant Quantum Monte Carlo computation for 2D boson-fermion coupling system

Research Interests

- Many-Body System: Analytical and numerical computation based on Quantum Monte Carlo, Tensor Networks, Exact Diagonalization and information theory.
- Explaining experiment-related novel phenomena in strongly correlated materials.

Selected Publications

Bilinear correlations in Fluctuating Gaussian States with anti-unitary symmetries *2026*
X Zhang
arXiv preprint arXiv:2609.01724

How a bilayer Nickelate superconducts: a Quantum Monte Carlo study *2025*
X Zhang
arXiv preprint arXiv:2512.09025

Constraints from anti-unitary symmetries on phase diagrams of sign problem-free models *2025*

X Zhang, N Bultinck

Physical Review B 112 (12), 125110

Magnetic Bloch states at integer flux quanta induced by super-moiré potential in graphene aligned with twisted boron nitride *2025*

Y Ma, M Huang, X Zhang, ...

Nature Communications 16 (1), 1860

Emergent conformal symmetry at the multicritical point of (2+1)D SO(5) model with Wess-Zumino-Witten term on a sphere *2024*

BB Chen, X Zhang, ZY Meng

Physical Review B 110 (12), 125153

Phases of (2+1)D SO(5) Nonlinear Sigma Model with a Topological Term on a Sphere: Multicritical Point and Disorder Phase *2024*

BB Chen, X Zhang, Y Wang, K Sun, ZY Meng

Physical Review Letters 132 (24), 246503

Integral algorithm of exponential observables for interacting fermions in quantum Monte Carlo simulations *2024*

X Zhang, G Pan, BB Chen, K Sun, ZY Meng

Physical Review B 109 (20), 205147

Intrinsic nonlinear Hall effect and gate-switchable Berry curvature sliding in twisted bilayer graphene *2023*

M Huang, Z Wu, X Zhang, ...

Physical Review Letters 131 (6), 066301

Polynomial sign problem and topological Mott insulator in twisted bilayer graphene *2023*

X Zhang, G Pan, BB Chen, H Li, K Sun, ZY Meng

Physical Review B 107 (24), L241105

Superconductivity and bosonic fluid emerging from moiré flat bands *2022*

X Zhang, K Sun, H Li, G Pan, ZY Meng

Physical Review B 106 (18), 184517

Fermion sign bounds theory in quantum Monte Carlo simulation *2022*

X Zhang, G Pan, XY Xu, ZY Meng

Physical Review B 106 (3), 035121

Momentum space quantum monte carlo on twisted bilayer graphene *2021*

X Zhang, G Pan, Y Zhang, J Kang, ZY Meng

Chinese Physics Letters 38 (7), 077305

Honor & Award

IOP Publishing China Top Cited Paper Award in Physics *2024*

Chinese Physical Society & IOP Publishing

QuantEmX Scientist Exchange Award *2022*

Institute for Complex Adaptive Matter & Gordon and Betty Moore Foundation, USA

Merit Student *2018*

Nankai University, Tianjin, China

Integrated Second-class Scholarship *2018*
Nankai University, Tianjin, China

Dalian Institute of Chemical Physics Scholarship *Jun. 2018*
Dalian Institute of Chemical Physics, Chinese Academy of Science, Liaoning, China

Jici Yan Class Scholarship *2015 - 2017*
Institute of Physics, Chinese Academy of Science, Beijing, China

Integrated First-class Scholarship *2016 & 2017*
Nankai University, Tianjin, China